Calibrated Reliance on AI Writing Assistants: Effects of Confidence Cues on Accepting Suggestions

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Sample paper written for the ko-paper README. All data are illustrative.

Abstract — People increasingly draft text with AI writing assistants, yet they often accept suggestions without checking them. We study whether showing a confidence cue next to each suggestion helps users rely on the assistant only when it is likely to be right. In a within-subjects study with 48 participants who edited short reports, confidence cues reduced the acceptance of incorrect suggestions by 31% while leaving the acceptance of correct ones almost unchanged. Participants also reported trust that tracked the assistant's actual accuracy more closely. We discuss how interface cues can support appropriate reliance rather than simply increasing trust.

1 Introduction

Writing assistants built on large language models now suggest whole sentences while people type. These suggestions save time, but they also invite overreliance: users accept fluent text even when it contains factual errors. Prior work on automation argues that the goal is not maximal trust but *calibrated* trust, in which a person's reliance matches the system's real capability.

We ask a simple question: does a visible confidence cue change which suggestions people accept? We contribute (1) a controlled study of confidence cues in a writing task, (2) a measure of calibration that combines acceptance and accuracy, and (3) design guidance for assistants that communicate uncertainty.

2 Method

Each participant edited twelve short reports. For every report the assistant offered four suggestions, one of which contained a planted error. In the *cue* condition, each suggestion carried a high, medium, or low confidence label; in the *baseline* condition, no label was shown. Condition order was counterbalanced.

We define the calibration score C of a participant as the difference between the acceptance rate of correct suggestions and the acceptance rate of incorrect ones:

$$C = P(\text{accept} \mid \text{correct}) - P(\text{accept} \mid \text{incorrect}) \quad (1)$$

A score of 1 means the participant accepted every correct suggestion and rejected every incorrect one; a score of 0 means acceptance did not depend on correctness.

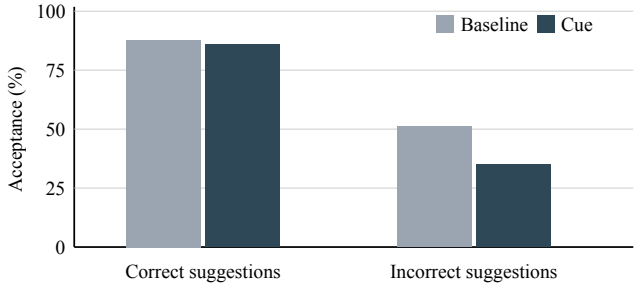


Figure 1: Acceptance rates by suggestion correctness. Confidence cues mainly lowered the acceptance of incorrect suggestions.

3 Results

Table 1 summarizes the main measures. With confidence cues, participants accepted far fewer incorrect suggestions, while acceptance of correct suggestions dropped only slightly. As a result, the mean calibration score rose from 0.37 to 0.51.

Table 1: Mean measures per condition (N = 48).

Measure	Baseline	Cue	Change
Accepted, correct (%)	87.8	85.9	−1.9
Accepted, incorrect (%)	51.0	35.2	−15.8
Calibration score C	0.37	0.51	+0.14
Task time (min)	6.4	6.9	+0.5
Self-reported trust (1–7)	5.6	5.1	−0.5

Editing took about half a minute longer with cues, which participants attributed to checking low-confidence suggestions more carefully. Self-reported trust decreased slightly, but it correlated more strongly with the assistant's accuracy on each report.

4 Discussion

Confidence cues did not make people trust the assistant more; they made people trust it more selectively. Designers should therefore evaluate assistants by calibration rather than

by acceptance alone. A limitation is that our cues were perfectly informative; real confidence estimates are noisy, and poorly calibrated cues could mislead users.

References

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